

Training Configuration and Summary:

Sequencing:

- timesteps: 60
- batch_size: 128

Model Hyperparameters:

- min_units: 128
- max_units: 256
- units_step: 64
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh']
- optimizers: ['adam', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

Directories:

- directory: my_dir
- project_name: lstm_hyperparameter_tuning

LSTM Training Report

Report Generated: 2025-04-09_13-22-20

Training Start: 2025-04-09 12:31:20

Training End: 2025-04-09 13:08:31

Training Duration: 2231.01 seconds

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: adam

Dropout Rate: 0.20

Units per Layer:

Layer 1: 256 units, Layer 2: 256 units

Performance Metrics:

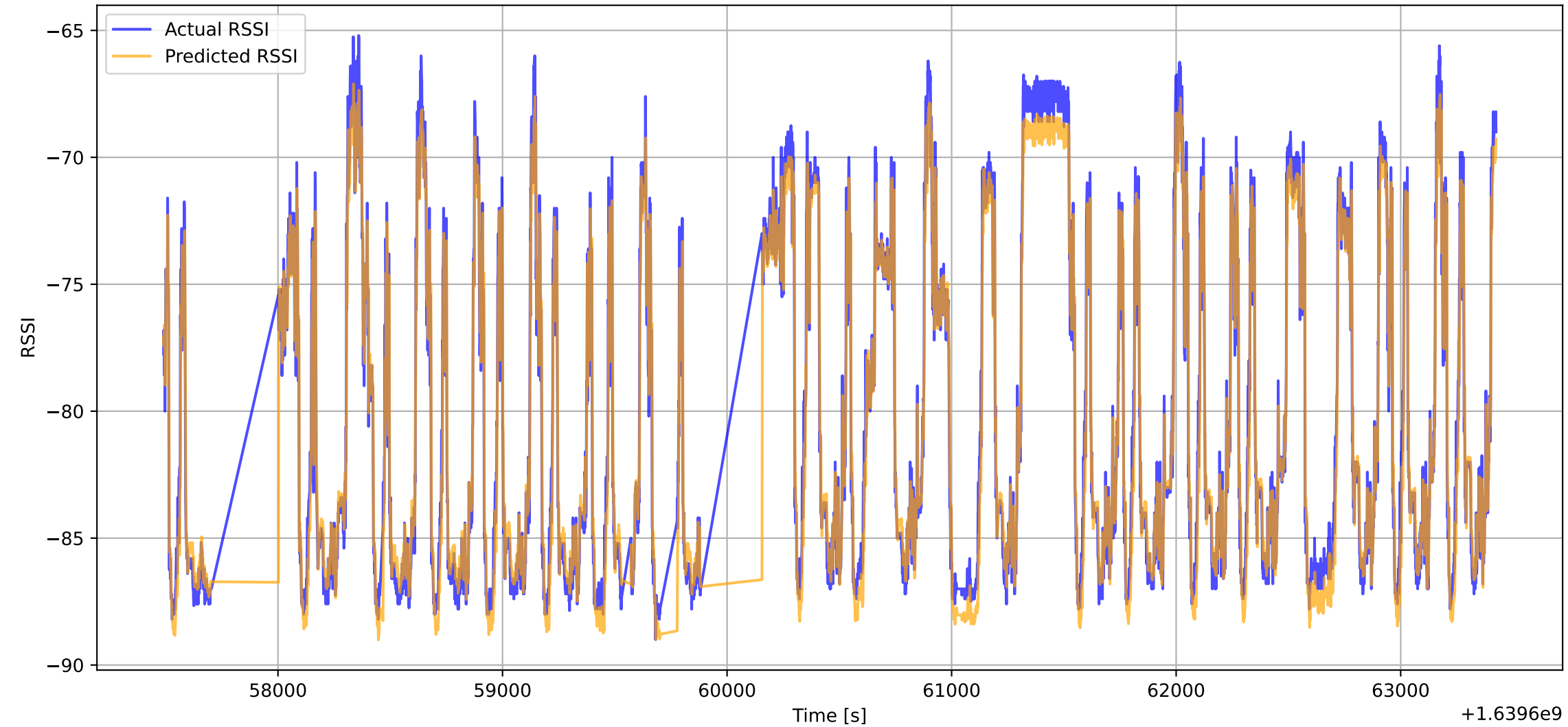
MAE: 1.0341

MSE: 2.0154

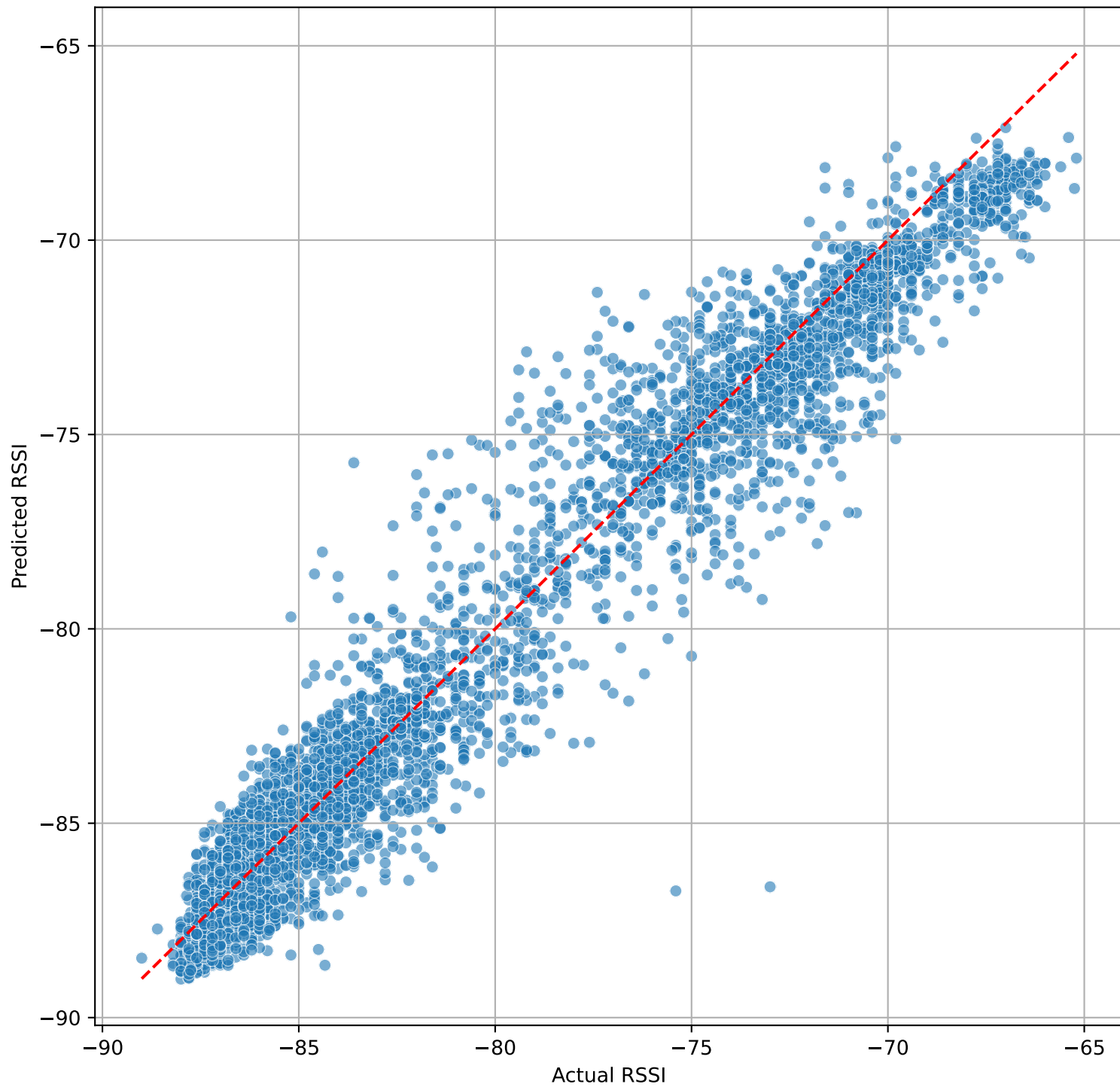
RMSE: 1.4197

R2 Score: 0.9542

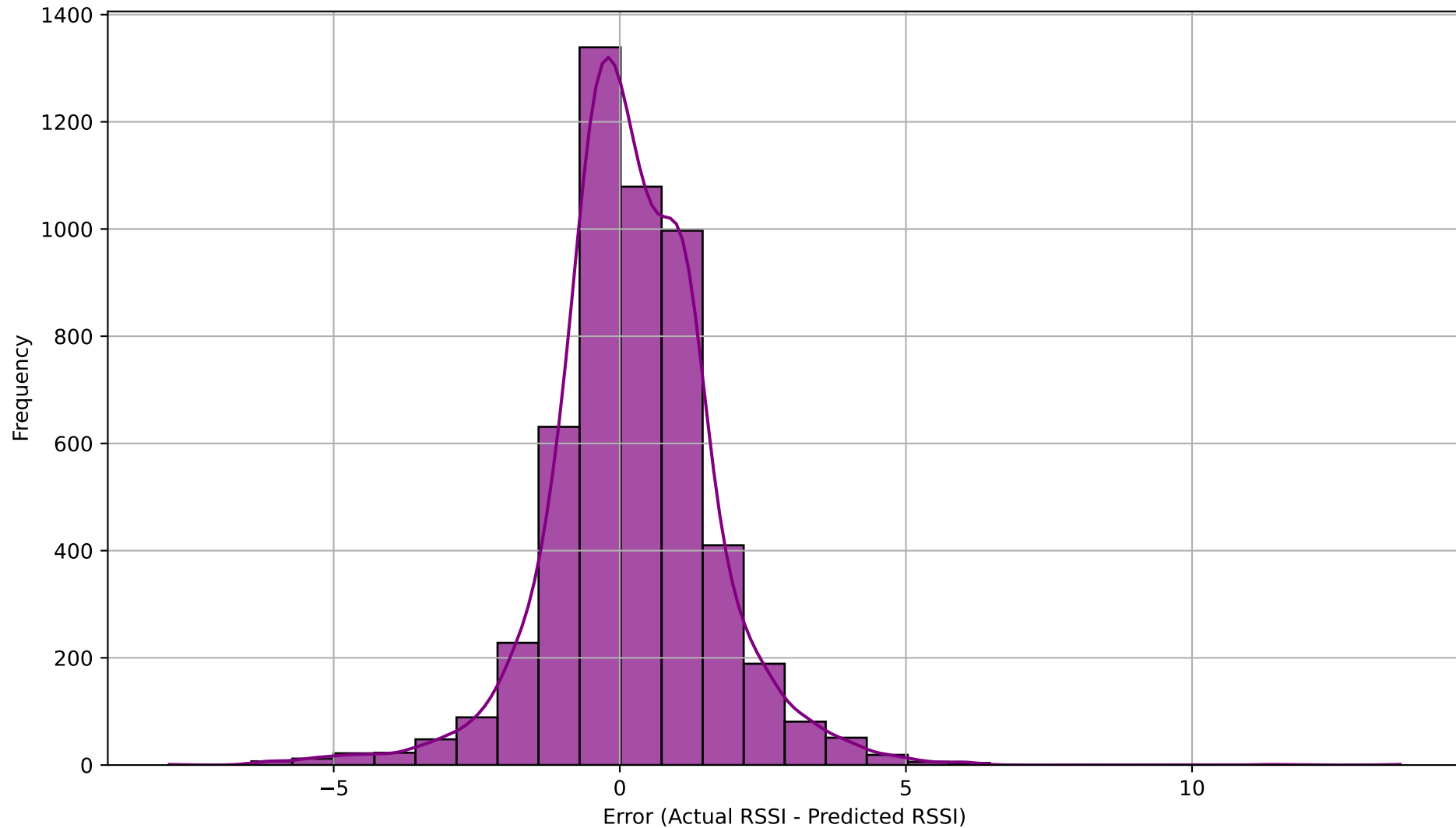
Actual vs Predicted RSSI Over Time



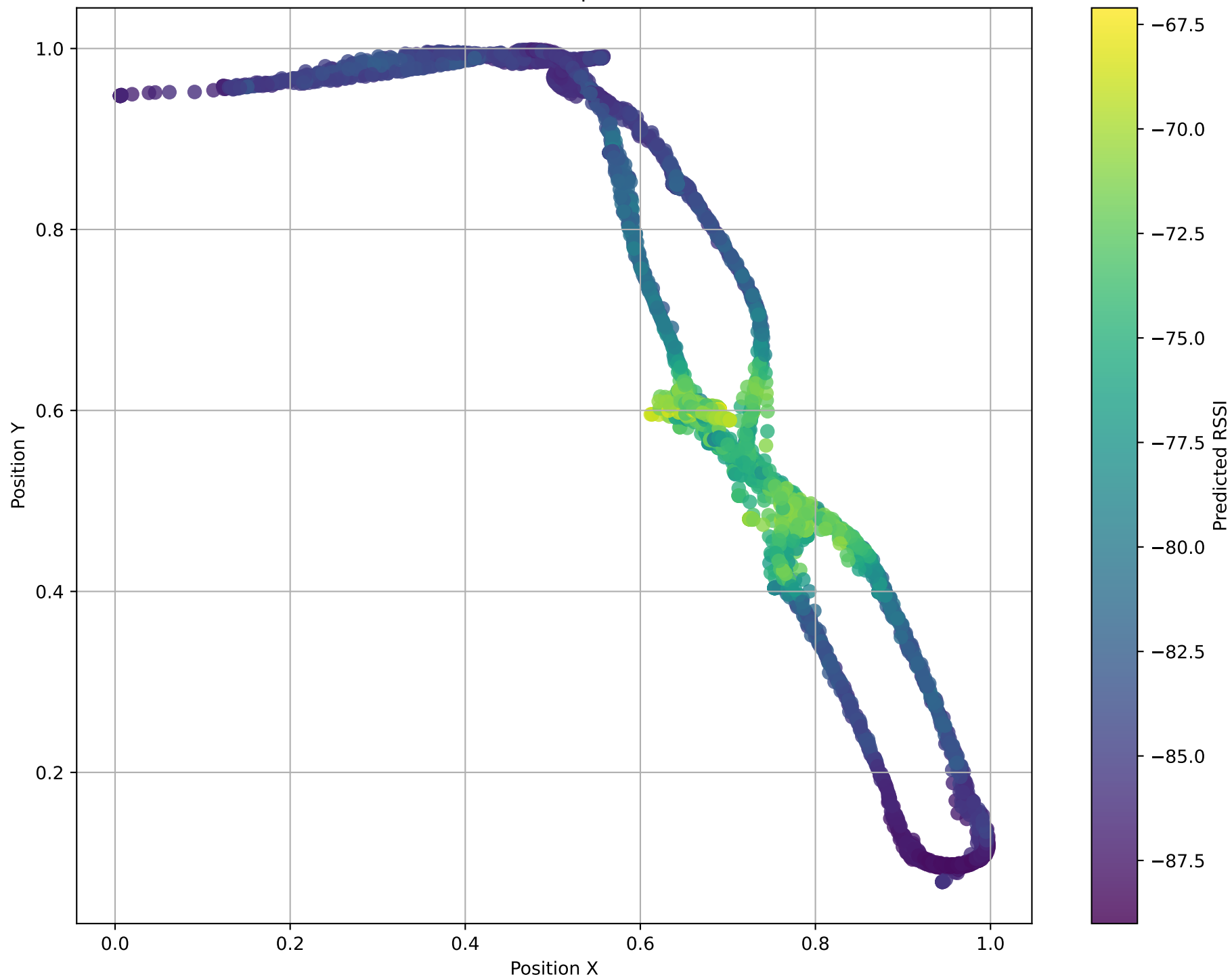
Actual vs Predicted RSSI



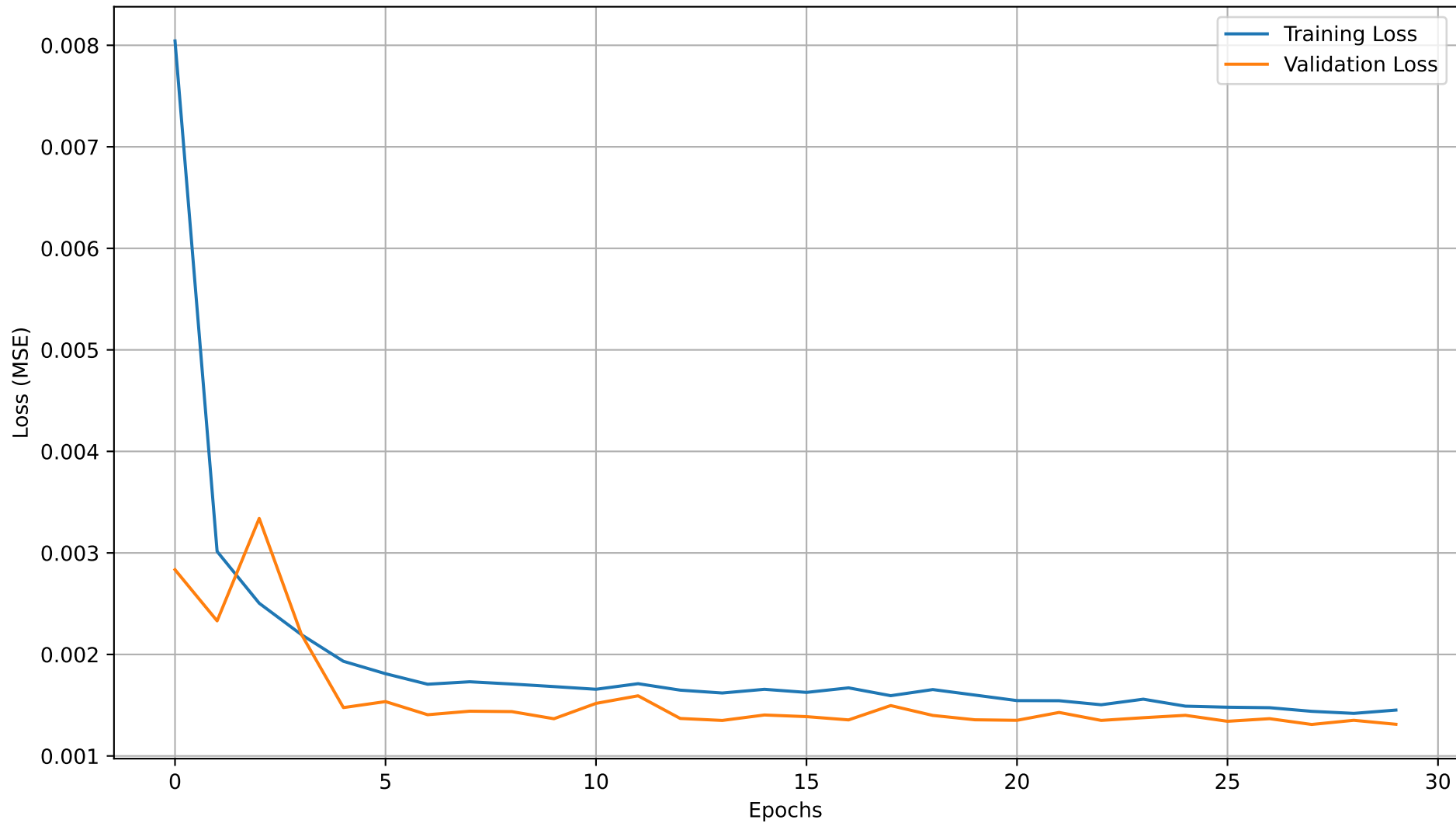
Distribution of Prediction Errors



Predicted RSSI in Spatial Context



Training and Validation Loss Over Epochs



Training and Validation MAE Over Epochs

